

The Unified Hierarchical Reasoning Controller

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Abstract

This paper presents the Unified Hierarchical Reasoning Controller (UHRC), a novel framework designed to enhance the reasoning capabilities of artificial intelligence systems. The UHRC integrates hierarchical reasoning processes, allowing for more efficient and effective decision-making in complex environments. We evaluate the performance of the UHRC through a series of experiments, demonstrating its superiority over traditional reasoning controllers in various tasks. The results indicate that the UHRC can significantly improve the adaptability and robustness of AI systems, making it a promising approach for future research in artificial intelligence and machine learning.

1 Introduction

The autonomous control of complex nonlinear systems remains one of the most consequential open problems in control engineering (Iqbal et al. 2017; Z. Liu et al. 2015; Daoutidis, Megan, and W. Tang 2023). Systems such as unmanned aerial vehicles (UAVs), autonomous ground vehicles, and high-degree-of-freedom robotic manipulators are characterised by high-dimensional state spaces, nonlinear inter-state couplings, actuator saturation, and significant parametric uncertainty (Nasser Sadati, Rahmani, and Saif 2015; Ameli and Anubi 2021; Iqbal et al. 2017; Slotine and W. Li 1991). The fundamental difficulty is that the dynamical regimes governing long-horizon navigation and those governing fast stabilisation operate on incommensurable timescales, differ in their sensitivity to model error, and impose competing demands on the same actuator plant (Liang et al. 2018; Abdelmaksoud, Mailah, and Abdallah 2020). Designing a controller that simultaneously addresses long-horizon planning and fast reactive stabilisation across these coupled dynamics is fundamentally a multi-scale reasoning problem, and no single architecture has proven universally adequate across the competing demands of stability, adaptability, and computational tractability (Rojas, Yepes, and Garcia 2025; Goodwin, Graebe, and Salgado 2001).

Classical hierarchical control has long addressed this multi-scale challenge through explicit structural decomposition: a high-level supervisory layer generates strategic plans and setpoints while a low-level regulatory layer executes fast stabilisation and trajectory tracking (Singh and Titli 1978; Nasser Sadati, Rahmani, and Saif 2015; Scattolini and Colaneri 2007; Mesarovic, Macko, and Takahara 1970; Wilson 1979). The theoretical underpinnings of this separation trace to the multilevel systems framework of (Mesarovic,

Macko, and Takahara 1970) and the hierarchical decomposition methods formalised in (Singh and Titli 1978; Siljak 2013), which established that large-scale optimisation problems can be rendered tractable by decomposing them into coordinated subproblems across levels. This architecture has achieved sustained success across aerospace, process control, and robotics, benefiting from well-established stability theory, modular implementation, and interpretable failure modes (Sanders et al. 1998; X.-F. Wang, Jiang, and W.-H. Chen 2024; M. Liu, Tan, and Padois 2016; Bahnasawi, Al-Fuhaid, and M. Mahmoud 1990; Roberts and Becerra 2001). The quadrotor UAV exemplifies this paradigm directly: the natural timescale separation between slow translational dynamics and fast rotational dynamics motivates a cascaded structure in which outer-loop position planners generate attitude references consumed by inner-loop attitude regulators (Hoffmann et al. 2007; Liang et al. 2018; Bouabdallah, Noth, and Siegwart 2004; Nagaty et al. 2013; Abdelhay and Zakriti 2019). Model predictive control has provided a principled realisation of the supervisory layer in such architectures, enabling constraint-aware planning with formal stability guarantees at the coordination level (Scattolini and Colaneri 2007; X.-F. Wang, Jiang, and W.-H. Chen 2024; Torrente et al. 2021).

However, the fixed interface between layers introduces a structural information bottleneck that becomes consequential as task complexity grows (Kolchinsky, Tracey, and Wolpert 2019; S. Li et al. 2023; Abdelmaksoud, Mailah, and Abdallah 2020). The high-level planner typically transmits heavily compressed information—ordinarily a kinematic waypoint sequence—to the low-level regulator, discarding contextual information regarding obstacle proximity, actuator feasibility, and local geometric constraints that may be critical for downstream control quality (Zhu, Ferrari-Trecate, and Geroliminis 2024; Zhou et al. 2017; W. Liu, Zheng, and Cai 2013). Conversely, the low-level regulator executes reference trajectories without access to the global context that motivated them, rendering local disturbance responses globally suboptimal (Nguyen, Lowenberg, and Neild 2021; Daoutidis, W. Tang, and Allman 2019). Fixed rule-based supervisory frameworks also become brittle when the system encounters scenarios not anticipated at design time (N. Sadati and Ramezani 2008; Dave et al. 2023). These limitations are not incidental but structural: they arise from the requirement that the planner and regulator communicate through a pre-specified, fixed-dimensional interface whose information content is determined at design time rather than adapted online to task demands (Kolchinsky, Tracey, and Wolpert 2019; Zhang, Lu, and Gao 2015).

Learning-based approaches have emerged as a principled alternative. End-to-end reinforcement learning (RL) enables joint optimisation of planning and regulation without manual interface specification (AlMahamid and Grolinger 2021; Levine et al. 2016; Kiran et al. 2022; Shakya, Pillai, and Chakrabarty 2023). Behavioural cloning provides a supervised alternative that sidesteps the credit-assignment challenges of RL by learning directly from expert demonstrations, and has seen substantial success in autonomous driving and manipulation (Levine et al. 2016; Le Mero et al. 2022; Ly and Akhloufi 2021; Block, Foster, and Misra 2024). However, flat end-to-end policies are notoriously sample-inefficient, sensitive to reward shaping, and do not naturally maintain the multi-scale temporal structure that long-horizon UAV navigation demands: the policy allocates identical computational depth regardless of whether the task requires strategic deliberation or reactive stabilisation (AlMahamid and Grolinger 2021; Agyei, Sarhadi, and Polani 2025). Deep reinforcement learning approaches applied directly to quadrotor control have demonstrated impressive agility in simulation (Trad et al. 2024; Saviolo and Loianno 2023), but the sim-to-real gap, sample complexity, and absence of formal stability guarantees remain

open challenges (Dai et al. 2021; Agyei, Sarhadi, and Polani 2025). Hybrid neuro-symbolic and fuzzy-integrated controllers partially address this brittleness by pairing learned components with structured decomposition (H. H. Tang and Ahmad 2024; T. A. Mahmoud et al. 2024; Tran et al. 2021; Baheri and Alm 2025), yet the supervisory-regulatory interface and its dimensionality remain manually prescribed. The boundary between planning and regulation is fixed by the designer, not learned from data (Borri, Pola, and Benedetto 2019; Yu et al. 2023).

Recent advances in neural reasoning architectures offer a substantially different approach to this problem. Test-time scaling methods that iterate computations in latent space (Geiping et al. 2025; H. Li et al. 2025) and recurrent architectures that adapt computational depth to task difficulty (Geiping et al. 2025; Z.-Z. Li et al. 2025) suggest that the depth of processing required for a given decision can be made emergent rather than fixed. The Hierarchical Reasoning Model (HRM), introduced by (G. Wang et al. 2025), is a two-tier recurrent architecture comprising a high-level module responsible for slow, abstract planning and a low-level module handling rapid, detailed computation, coupled through a shared carry state that enables adaptive allocation of computational depth. Trained on as few as 1,000 examples without pre-training or chain-of-thought supervision, the HRM achieves near-perfect accuracy on complex Sudoku problems and optimal pathfinding in large mazes—tasks on which state-of-the-art large language model approaches fail entirely (G. Wang et al. 2025; Huang and Chang 2023; Feng et al. 2025). Crucially, the hierarchical role specialisation between modules emerges through training rather than being architecturally prescribed, mirroring the representational dimensionality hierarchy observed across cortical regions in the mammalian brain (Badre and Nee 2018; Huntenburg, Bazin, and Margulies 2018; Murray et al. 2014), where higher-order areas operate over longer intrinsic timescales and higher-dimensional representations than primary sensory regions. Building on this, (Jin et al. 2025) demonstrated that decoupling planning from execution in a hierarchical framework yields measurable gains in multi-step reasoning tasks, further motivating the application of this design principle to closed-loop control. The hierarchical neuro-symbolic decision transformer of (Baheri and Alm 2025) similarly demonstrates that coupling a symbolic planner with a learned low-level policy through a structured interface outperforms both purely symbolic and purely neural baselines on sequential decision tasks, providing additional evidence that the hierarchical planning-execution separation confers a genuine computational advantage beyond what either component achieves in isolation.

Nevertheless, all demonstrated HRM applications to date have been confined to discrete, turn-based domains. Whether the same unified reasoning mechanism can function as a closed-loop controller for a continuous, real-time nonlinear dynamical systems where control outputs must be physically continuous, observations arrive as dense sensory streams, and temporal context must persist across thousands of control cycles remains entirely unexplored in the literature. Extending HRMs to such settings requires non-trivial architectural adaptations: the per-problem hidden state must become a persistent rolling buffer, discrete token outputs must be replaced by continuous wrench commands, and the supervised training signal must be derived from a physically meaningful expert policy rather than symbolic ground truth labels. Localisation and mapping, which are prerequisite for any meaningful planning computation, must be supplied by an onboard SLAM pipeline (Gupta and Fernando 2022; Milić et al. 2021; Munguia, Trujillo, and Grau 2025; W. Chen et al. 2022) whose outputs form part of the continuous observation stream consumed by the controller.

This paper addresses that gap by proposing and evaluating the *Unified Hierarchical Reasoning Controller* (UHRC) for the autonomous navigation of a six-degree-of-freedom quadrotor UAV, constituting the first systematic application of the HRM architecture to a continuous nonlinear closed-loop control system. The UHRC adapts the HRM’s two-level reasoning structure into a recurrent transformer controller trained via behavioural cloning (Levine et al. 2016; Le Mero et al. 2022; Block, Foster, and Misra 2024) on expert trajectory data generated by an A* path planner (LaValle 2006; Qin et al. 2023) across 500 simulated episodes. A persistent 16-slot rolling carry buffer replaces the HRM’s per-problem hidden state, providing bounded temporal context without unbounded memory growth. The architecture is evaluated against a classical hierarchical baseline incorporating SLAM-based localisation (Gupta and Fernando 2022; Milijas et al. 2021), A* path planning, and cascaded PID control (Bouabdallah, Noth, and Siegwart 2004; Ren et al. 2016; Abdelhay and Zakriti 2019; Lopez-Sanchez and Moreno-Valenzuela 2023) across five structured test cases of increasing environmental complexity.

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The primary contributions of this work are as follows:

1. The first empirical demonstration of an HRM-based architecture operating as a closed-loop controller for a continuous nonlinear dynamic system, establishing the functional feasibility of unified hierarchical reasoning in the control domain.
2. A UHRC architecture incorporating dual-stream fusion of kinematic and LiDAR observations, a rolling carry buffer for persistent temporal context, Rotary Positional Encoding (RoPE) over the carry sequence, and a gated subgoal injection mechanism coupling H-module velocity planning to L-module wrench computation.
3. A behavioural cloning training pipeline employing a versatile episode distribution, A*-generated expert demonstrations with injected perturbations and a structured loss combining weighted action imitation, velocity subgoal supervision, and cosine alignment.

4. A rigorous comparative empirical evaluation quantifying the performance robustness trade-off between unified learned hierarchy and classical explicit decomposition across open-field transit, constrained corridor navigation, gap threading, dynamic obstacle avoidance, and parametric model uncertainty.

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